

Krishna Birla

+91 9588620228 | krishnabirla336@gmail.com | [in krishnabirla](https://www.linkedin.com/in/krishnabirla) | [G Krishna9588](https://github.com/Krishna9588) | Pune, MH

Experience

Jr. ML Associate — ProPlus Data

Jun 2025 - Present

- Led the **backend** development, **AI integration**, and **MongoDB** Atlas setup to manage data and build a custom RAG system in **Nexora**. Resulting in an AI that answers queries based exactly on user purchases.
- Scraped company specific data to **fine tune** a Llama 3 model and built an **RAG system** around it, giving the **Nexora AI** real time web access to answer queries based exactly on what the user purchased.
- Replaced the External API in the Decision Logic Pipeline with local language & reasoning models to reach a concurrency from **1 to 6**, which cut costs and made processing **4x** faster without affecting the result.

Backend Developer Intern — Aroma Brand Solutions

Jan 2025 - Mar 2025

- Architected and deployed a high concurrency Python/Flask backend infrastructure utilizing advanced **session state management** to optimize crawlability and global SEO performance.
- Implemented a language model based document retrieval system to automate content creation, generating precise FAQs, questions and answers, and clear topic summaries.

Technical Skills

Languages: Python, JavaScript, MySQL

AI and GenAI: PyTorch, TensorFlow, Hugging Face, LangChain, RAG, Local LLMs Fine-Tuning, Agentic Workflows, Vector Embeddings, Distributed Systems, API Development, Data Processing, Web Scraping

Cloud and Databases: AWS, GCP, MongoDB, SQLite, Supabase, ChromaDB

Testing and Automation: Pytest, Selenium, Playwright, Scrapy, AutoGen, Langfuse.

Tools and Version Control: GitHub, Docker, Pytest, Selenium, Playwright.

Projects

Founders Intelligence | Python, FastAPI, LangChain, Gemini API, Apify, MongoDB Atlas

| [Demo](#) | [Link](#)

- Built a pipeline of **5 connected AI agents** that replace manual research roles by automatically turning raw competitor data, news, and social media signals into formatted **product insights**.
- Created **automated scraping** connectors to extract high signal information from play stores, user conversations, video transcripts, documents, and internal notes for deep document analysis.
- Designed a domain aware **RAG framework** and **chat interface** using the API and MongoDB Atlas to analyze these multi source datasets, giving instant answers backed by actual evidence.

Verification Engine | Python, LLMs, Docker, Selenium, Playwright, Pandas

| [Project Link](#)

- Built a **data validation system** that extracts information from sources and files to confirm data accuracy and prevent false entries from entering company databases.
- Deployed **language models** inside **Docker containers** to handle large scale batch processing while using Selenium, Playwright, and Pytest to run automated tests and monitor data quality metrics in real time.

Archiscan AI | Python, GenAI, OpenCV

| [Demo](#) | [Link](#)

- Developed a **Gen AI** platform using Python and OpenCV that automatically maps land boundaries and generates structural blueprints while verifying compliance with Indian laws and national regulations.
- An **image processing** and **neural network architecture** to replace the manual 5-day drawing process, with a permanent 1-day solution that supports multiple file outputs and cuts the 50k Rs per project cost.

Education

B.E. in Artificial Intelligence and Machine Learning

CGPA: **8.30 out of 10**

ISBM College of Engineering, Pune.

2022 - 2026

Certificates & Publication

- IEEE** | Architectural Design by Blueprint Generation using GenAI | **Research Paper** May 2026 | [Link](#)
- Oracle** | Certified **Generative AI** Professional | **Exam Certificate** Oct 2025 | [Link](#)
- IJSRET** | Landslide Prediction Using ML and GIS-Based Approach | **Research Paper** May 2025 | [Link](#)
- DataKind** Financial Inclusion | **Open Source Contribution** Apr 2025 | [Link](#)